

Tutorial-1

EM-613: Mathematical Models for Data Science

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Maximum marks: 12

Please attempt all the questions. Submit only one file name `t1_12345678.zip` which contains all the script files required by the questions below.

1. Write a MATLAB script file name `p1.m` that does the following:
 - (a) Creates a 3×4 matrix, $\mathbf{A}$ with random decimal values, $x \in [0, 1]$. Display the output on the command line. (1)
 - (b) Calculate the $\mathbf{A}\mathbf{A}^T$ and display the output on the command line. (1)
2. Write a MATLAB file `p2.m` that determines whether the discriminant of the quadratic equation $ax^2 + bx + c = 0$ is positive, negative or zero. The output is displayed onto the command window. Calculate the roots of the quadratic equation and verify your answer using the inbuilt MATLAB functions `poly` and `roots`. (4)
3. Write a MATLAB file `p3.m` that iteratively solves the equation $x = \ln(x)$ and prints the number of iterations along with the solution upto 10 decimal places. Verify your answer using the built-in function `fsolve`. (3)
4. Write a MATLAB file `p4.m` that allows for a graphical solution of the equation $x = \cos(x)$ in the range $x \in [0, \frac{\pi}{2}]$. Format the plot, along with legends, so that it is legible. Save the figure using the `print` command as `p4.png`. (3)